Milestone 2: EDA Dashboard

QM 2023 Capstone Project

Due: Friday, Week 10 (March 27, 2026) by 11:59 PM **Points:** 50 (25% of capstone grade) **Format:** Team submission via shared GitHub repository (`main` branch)

Overview

Milestone 2 transitions from **data engineering (M1)** to **exploratory data analysis**. Your goal is to uncover patterns, correlations, and relationships in your data that will guide your econometric specifications in M3.

Real-world context: Before running regressions, analysts must understand their data. What variables correlate? At what lags? Are there subgroups with different sensitivities? EDA is where hypotheses are formed, not tested.

Success criterion: Your visualizations should be **publication-ready** (titles, labels, legends, captions) and every plot should tell a story that informs your M3 models.

> **Note:** Guidance and short snippets are provided for each required plot. Adapt patterns to your dataset and variable names — do not copy full solutions.

Dataset Adaptation Note

Alternative Dataset Teams: The required plots below use REIT examples, but the **concepts** apply to all panel datasets: - **Correlation heatmap** → Works for any dataset (outcome + drivers + controls) - **Time series** → Plot your outcome variable over time - **Dual-axis** → Overlay outcome + key driver (e.g., returns + rates, prices + sentiment) - **Lagged effects** → Test which lag structure is optimal for your driver variable - **Group analysis (Plots 5-6)** → **CONDITIONAL**: If your dataset has groups (sectors, regions, asset types), complete these plots. If not, replace with alternatives (see "Dataset Without Groups" section below). - **Scatter plots** → Bivariate relationships (outcome vs. control variables) - **Decomposition** → Trend + seasonal + residual components

Translation Guidance: See `Dataset-Translation-Examples.md` for worked examples translating REIT plots to Crypto, Housing, and Macro datasets.

> **Open Data Catalog Teams:** If your dataset comes from the Open Data catalog, the same plot types apply. Replace "REIT returns" with your outcome variable, "FEDFUNDS" with your driver variable, and "sectors" with whatever grouping variable exists in your data (if any). If you don't have natural groups, see the "Alternatives for Datasets Without Groups" section below.

Learning Objectives

By completing M2, you will:

1. **Create publication-quality visualizations** (titles, labels, legends, captions)
2. **Identify correlation patterns** between outcome and driver variables
3. **Determine optimal lag structures** for time-series relationships
4. **Detect group heterogeneity** (if applicable to your dataset)

5. **Formulate testable hypotheses** for M3 econometric models
6. **Diagnose data quality issues** (outliers, missing values, heteroskedasticity)
7. **Communicate visual insights** through economic interpretation

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Deliverables

1. Jupyter Notebook: `capstone_eda.ipynb`

Requirements: - Runs from top to bottom without errors (Restart Kernel → Run All) - Minimum **8 required visualizations** (see below) - Every plot has: - Descriptive title - Axis labels with units - Legend (if multiple series) - Caption explaining insight + economic interpretation - Saved to `results/figures/` as PNG files (300 DPI). Use `FIGURES_DIR` from config_paths.

Structure: Section 1: Imports + data loading (from config_paths, load M1 output) → Section 2: Summary statistics → Section 3: Correlation analysis → Section 4: Time series → Section 5: Lagged effect analysis → Section 6: Group analysis (if applicable) → Section 7: Factor/control relationships → Section 8: Time series decomposition

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2. Required Visualizations (8 minimum)

Plot 1: Correlation Heatmap (REQUIRED)

Purpose: Identify which variables are strongly correlated with your outcome variable.

REIT Example Variables: `ret`, `fedfunds`, `mortgage30us`, `cpiaucsl`, `unrate`, `mom`, `qlty`, `size`

Generic Pattern: Include your `[OUTCOME]`, `[DRIVER]`, and `[CONTROL]` variables.

Interpretation Guidance: - What is the strongest correlation with your outcome? (Will likely be your main driver for M3) - Are any control variables highly correlated with each other? (Multicollinearity warning for M3) - Do correlations match your theoretical expectations?

Guidance: Use `sns.heatmap()` on your correlation matrix. Include outcome, driver, and controls. Set `sns.set_style("whitegrid")` and save to `FIGURES_DIR`. Key snippet: `corr_matrix = data[vars_to_plot].corr()`.

Caption Template: > "The heatmap reveals a [strong/moderate/weak] negative correlation (r = X.XX) between [OUTCOME] and [DRIVER]. [Explain why this makes economic sense]. [CONTROL1] shows [pattern], suggesting [implication for M3]."

Plot 2: Time Series of Outcome Variable (REQUIRED)

Purpose: Visualize trends, volatility, and outlier periods.

Generic Pattern: Plot `[OUTCOME]` on Y-axis, `[TIME]` on X-axis.

REIT Example: Average REIT returns over time (2015-2024).

Crypto Example: Bitcoin returns over time (2020-2024).

Housing Example: Median home price growth rate over time (2010-2024).

Interpretation Guidance: - Identify crisis periods (e.g., March 2020 COVID crash, 2022 rate hikes) - Note volatility clustering (periods of high vs. low variance) - Detect secular trends (persistent upward/downward movement)

Plot 3: Dual-Axis Plot (Outcome vs. Driver) (REQUIRED)

Purpose: Visualize co-movement between outcome and key driver variable.

Generic Pattern: Left Y-axis = `[OUTCOME]`, Right Y-axis = `[DRIVER]`, shared X-axis = `[TIME]`.

REIT Example: REIT returns (left) vs. Federal Funds Rate (right).

Crypto Example: Token returns (left) vs. Regulatory Severity Index (right).

Housing Example: Home price growth (left) vs. Mortgage rates (right).

Interpretation Guidance: - Do the two series move together (positive correlation) or opposite (negative)? - Is there a visible lag (driver moves first, outcome follows)? - Are there periods where the relationship breaks down?

Plot 4: Lagged Effect Analysis (REQUIRED)

Purpose: Determine at what lag the driver variable most strongly correlates with the outcome.

Method: Test lags 0, 1, 2, 3, 6, 12 (adjust for your frequency: months vs. days). Use `data.groupby('[entity_id]')['[driver]'].shift(lag)` to create lagged driver—**critical:** group by entity to avoid cross-entity leakage. Plot bar chart of correlation vs. lag. Identify optimal lag for M3.

Interpretation Guidance: - What is the optimal lag? (This becomes your lag specification in M3) - Does the relationship strengthen or weaken with longer lags? - What is the economic mechanism for this lag? (E.g., "REITs refinance debt after 2 months")

Plot 5: Group Box Plots (CONDITIONAL - See Alternatives Below)

When to Include: Your dataset has meaningful groups (sectors, regions, asset types, size quartiles).

Purpose: Compare outcome distributions across groups.

REIT Example: Box plots of returns by REIT sector (Retail, Office, Industrial, Residential).

Crypto Example: Box plots of returns by token type (DeFi, CEX, NFT, Stablecoin).

Housing Example: Box plots of price growth by region (Urban, Suburban, Rural).

Guidance: Use `data.boxplot(column='[outcome]', by='[group_variable]')` or seaborn boxplot. Remove default suptitle; add proper axis labels and units.

Interpretation Guidance: - Which group has the highest/lowest median outcome? - Are there outliers specific to certain groups? - Does variance differ across groups? (Heteroskedasticity warning for M3)

Plot 6: Group Sensitivity Analysis (CONDITIONAL - See Alternatives Below)

When to Include: Your dataset has groups AND you expect differential sensitivity to the driver.

Purpose: Segment groups based on sensitivity to the driver variable.

REIT Example: Sector-level correlation with FEDFUNDS; color-code sensitive ($r < -0.3$) vs. resilient ($r \geq -0.3$).

Crypto Example: Token type correlation with regulatory sentiment; identify most affected types.

Housing Example: Regional correlation with mortgage rates; identify rate-sensitive markets.

Guidance: Compute `group_sensitivity = data.groupby('[group_variable]').apply(lambda x: x['[outcome]'].corr(x['[driver]']))`. Plot horizontal bars; color-code by threshold (e.g., red if $r < -0.3$). Flag sensitive groups for M3 interaction terms.

Interpretation Guidance: - Which groups are most sensitive to the driver? - What economic mechanisms explain these differences? (E.g., "Retail REITs have higher leverage") - Should M3 include group × driver interaction terms?

Alternatives for Datasets Without Groups

If your dataset does **not** have natural grouping variables (e.g., single-entity time series, macro indicators), **replace Plots 5-6 with ONE of these alternatives:**

Alternative A: Time Period Subsample Analysis

Purpose: Compare relationship strength across different time periods (bull vs. bear, pre/post crisis). Split data by date; compute correlation per period; bar chart.

Alternative B: Rolling Correlation Analysis

Purpose: Show how the relationship changes over time. Use `data['[outcome]'].rolling(window=6).corr(data['[driver]'])` (adjust window for your frequency). Line plot over time.

Alternative C: Size Quartile Analysis

Purpose: Segment by size (market cap, volume, GDP). Use `pd.qcut(data['[size_variable]'], q=4)`; box plot outcome by quartile.

Plot 7: Factor/Control Variable Scatter Plots (REQUIRED)

Purpose: Visualize bivariate relationships between outcome and control variables.

Generic Pattern: Scatter plot of `[OUTCOME]` vs. `[CONTROL]` with regression line.

REIT Example: Returns vs. Momentum (MOM), Returns vs. Quality (QLTY).

Crypto Example: Returns vs. Trading Volume, Returns vs. Market Cap.

Housing Example: Price Growth vs. Employment Growth, Price Growth vs. New Construction.

Guidance: Use `sns.regplot()` for scatter + regression line. Add units to axis labels; save to FIGURES_DIR.

Plot 8: Time Series Decomposition (REQUIRED)

Purpose: Separate trend, seasonal, and residual components.

Guidance: Use `statsmodels.tsa.seasonal.seasonal_decompose`. Aggregate to time series first if panel data (`groupby('[time]')['[outcome]'].mean()`). Period: 12 for monthly, 365 for daily, 4 for quarterly. Plot observed, trend, seasonal, residual in 4-panel figure.

Interpretation Guidance: - Is there a clear upward/downward trend? - Are seasonal patterns strong? (If yes, include seasonal dummies in M3) - Are residuals white noise or do they show structure? (Structure → additional controls needed)

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3. Summary Markdown: `M2_EDA_summary.md`

Required sections: - **Key Findings (3-5 bullets):** Correlations, optimal lag, group sensitivity, outliers, control patterns—each with economic mechanism. - **Hypotheses for M3 (3+):** Hypothesis 1 (driver effect): claim, model spec, expected sign, mechanism. Hypothesis 2 (control premiums). Hypothesis 3 (group heterogeneity, if applicable). See `Dataset-Translation-Examples.md` for REIT/Crypto/Housing examples. - **Data Quality Flags:** Outlier periods, missing values, heteroskedasticity, multicollinearity—and planned M3 mitigations.

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4. AI Audit Appendix: `AI_AUDIT_APPENDIX.md`

Same requirements as M1. Document all AI use (ChatGPT, Copilot, etc.) with specific prompts, outputs, verification, and critique.

Technical Requirements

1. Jupyter Notebook Best Practices

Markdown Cells: - Use headings (`#`, `##`, `###`) to structure sections - Write narrative explanations before each visualization - Include captions after each plot explaining the insight

Code Cells: - One logical step per cell (easier to debug) - Use comments to explain complex code - Print output (summary statistics, row counts) to verify logic

Visualization Standards: - **Title:** Every plot must have a descriptive title - **Axis labels:** Include units (e.g., "Return (%)", "Month (YYYY-MM)") - **Legends:** Clearly label all lines or groups - **Color choices:** Use colorblind-friendly palettes (e.g., `seaborn.color_palette("colorblind")`) - **Font size:** Increase if needed for readability (`plt.rcParams['font.size'] = 12`)

Grading Rubric (50 points)

See `rubric.md` for detailed breakdown. Summary:

Component	Points	Criteria
Data Loading & Summary	10	Notebook runs without errors; data dimensions verified; summary stats presented
Visualization Quality	20	All 8 required plots present; publication-ready (titles, labels, legends); appropriate plot types
Analysis & Interpretation	15	Captions explain insights; economic intuition provided; patterns connected to theory
Hypothesis Formulation	5	Clear hypotheses for M3; testable and grounded in EDA findings

Total: 50 points

Common Pitfalls and How to Avoid Them

Pitfall 1: Plots Without Titles or Labels

Problem: Axes say `ret` and `fedfunds` with no explanation. **Solution:** Every plot needs: - Title explaining what is shown - X-axis label with units - Y-axis label with units - Legend if multiple series

Pitfall 2: No Economic Interpretation

Problem: Caption says "Returns and rates are negatively correlated" (just restates the visual). **Solution:** Explain *why* economically. "Rising rates increase financing costs for leveraged [entities], reducing profitability and depressing returns."

Pitfall 3: Ignoring Outliers

Problem: March 2020 COVID crash dominates all visualizations, obscuring patterns. **Solution:** Either: - Winsorize extreme values for visualization (document this) - Create separate crisis vs. normal period analysis - Add annotations to highlight outlier periods

Pitfall 4: Using Wrong Plot Type

Problem: Correlation matrix shown as a line plot (nonsensical). **Solution:** Match plot type to data: - Heatmap → Correlation matrices - Line plot → Time series - Box plot → Distributions across categories - Scatter plot → Bivariate relationships

Pitfall 5: Notebook Doesn't Run Top-to-Bottom

Problem: Cells depend on manual execution order; "Run All" fails. **Solution:** Restart kernel and run all cells before submission. Fix any errors.

Testing Checklist (Run Before Submission)

- [ ] **Restart kernel and run all:** Kernel → Restart & Run All. No errors?

- [ ] **All 8 visualizations present:** Check list of required plots (or alternatives if no groups)
- [ ] **Every plot has title, labels, legend**
- [ ] **Captions explain insights, not just describe visuals**
- [ ] **M2_EDA_summary.md complete:** Key findings, hypotheses, data quality flags
- [ ] **AI Audit Appendix submitted**
- [ ] **All files saved to results/figures/ and results/reports/:**
    - `capstone_eda.ipynb`
    - `M2_EDA_summary.md`
    - `AI_AUDIT_APPENDIX.md`
    - All PNG files from plots

Submission Instructions

Shared Team Repository Submission

1. **Work in your team's shared private GitHub repository** and submit by committing/pushing to `main`.
2. **Add your files:** `capstone_eda.ipynb`, `M2_EDA_summary.md`, `AI_AUDIT_APPENDIX.md`, `results/figures/M2_*.png`. Commit and push to `main`.
3. **Verify:** Check GitHub repo online; Jupyter notebook should render with all plots, and your latest commit should be on `main`.
4. **Auto-checks:** If repository checks run, use them to confirm notebook execution and file completeness.

Deadline: Friday, Week 10 (Mar 27) by 11:59 PM **Late penalty:** 10% per day, up to 3 days. After 3 days, no credit.

Resources and Support

Office Hours

- **Dr. Seagraves:** Monday & Wednesday, 3:00-5:00 PM (Helm 122-D)
- **Focus for M2:** Visualization design, economic interpretation, hypothesis formulation

Example Code

- **Starter notebook:** `starter/capstone_eda.ipynb` (incomplete template)
- **seaborn gallery:** Official examples
- **matplotlib customization:** User guide

Debugging Tips

- **Plot not showing?** Add `plt.show()` at end of cell.
- **Colors are ugly?** Use `seaborn.color_palette("colorblind")` for accessibility.
- **Lag analysis not working?** Use `groupby('[entity_id]').shift(lag)` to avoid cross-entity leakage.

Next Steps After M2

Your EDA insights will directly inform: - **Milestone 3 (Week 14):** Econometric model specifications (which lags? which interactions?) - **Milestone 4 (Week 16):** Investment memo narrative (which groups to recommend?)

Don't treat M2 as a checkbox exercise. The better your EDA, the stronger your M3 models and M4 memo will be.

Good luck! Great EDA is the bridge between data and insight.

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